



SOMAIYA

VIDYAVIHAR UNIVERSITY

K J Somaiya College of Engineering



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Syllabus

Honours Programme in

Software and Mobile Application Development

(Offered by Department of Computer Engineering)

From

Academic Year 2024-25

Revision 2

(Approved in Academic Council meeting dated 22/04/2024)



K J Somaiya College of Engineering, Mumbai-77

(Somaiya Vidyavihar University)

Honours' Degree Programme in Software and Mobile Application Development

Offered by Department of Computer Engineering

Introduction:

Last decade has witnessed Mobile handheld devices overtaking the conventional notion of desktop based computing, with iOS and Android emerging as de-facto operating systems. In recent years, due to their portability, computational capability at par with conventional computers Android and Apple iOS based devices are being preferred globally. This has resulted in a huge demand of professionals in the domain of Application development for Apple iOS and Android platforms.

Looking at the opportunities available in this field for employment and subsequent career growth it's the right time to start Honours programme in Software Application Development.

- Provision for two tracks viz, Apple iOS and Android platform. The Apple iOS authorized by Apple Training Centre covers Apple's state of the art development platform and related technologies. Whereas, the Android track will cover the Android specific development platforms e.g. Android-Studio and Flutter.
- Industry authorised program with complete support of coordination and guidance to blend and integrate Apple's curriculum including that of AI / ML, AR-VR so that the programme is always maintained up to date, in line with the technology advancements.
- Impart training to the Students on Apple iOS and help them excel in professional careers.
- Learning opportunities of advanced technologies on multi-purpose Mac Lab.
- Expand horizon of employment & entrepreneurship opportunities of students with Apple's internationally acceptable certification program.

Objectives:

- To train the students on industry certified curriculum so that they will be industry ready and thus excel in professional careers and contribute to social needs through application development.
- Students can pursue higher education, conduct research, demonstrate professionalism and ethics
- Students exhibit innovation, adaptability, team work, leadership and communication skills

Learning Outcomes of the Honours' Degree Programme:

At the successful completion of this programme, an Engineering Graduate will be able to:

- Understand the principles of user interface design and material design in the context of mobile application development.
- Develop proficiency in building cross-platform mobile applications using Swift, Kotlin, Flutter, and React Native frameworks.
- Implement responsive design techniques and utilize platform-specific features to ensure optimal user experience.
- Apply usability-testing methods and incorporate accessibility principles to evaluate and refine user interface designs for iOS and Android applications.

Eligibility criteria: Students should have passed First year of Computer Engineering or Computer Science and Business Systems (CSBS).

Assessment Methods: Evaluation is done by a variety of tools including Open book tests, MCQs (multiple-choice questions), Study of research papers, Internal Assessment tools and End Semester Examinations etc. Mini-Projects are offered in courses also to encourage project based learning among students.

Acronyms used in syllabus document	
Acronym	Definition
CA	Continuous Assessment
ESE	End Semester Exam
IA	Internal Assessment
O	Oral
P	Practical
P&O	Practical and Oral
TH	Theory
TUT	Tutorial
TW	Term work
ISE	In-semester Examination
CO	Course Outcome

Acronyms used in Course code e.g. 116HxxC301

Position of Digit	Acronym	Definition
1	2	SUV 2023 Second Revision
2	16	KJSCE
3	H	Honour Degree Program
4	05(xx)	Software and Application Development
5	C	Core Course
	L	Laboratory Course
	T	Tutorial



	P	Project Based Course
6	1/2/3/4	Semester Number
7	01/02/03--	Course Number

Proposed Credit Scheme

Course Code	Course Name	Teaching Scheme (Hrs.) TH – P – TUT	Total (Hrs.)	Credits Assigned TH – P – TUT	Total Credits	Suggested semester of Honours' degree
216H05C401	Fundamentals of Swift	3 – 0 – 0	03	3 – 0 – 0	03	IV
216H05L401	Application Development Lab I - Swift	0 – 2 – 0	02	0 – 1 – 0	01	IV
216H05C501	Mobile application development using Android – Kotlin / Flutter	3 – 0 – 0	03	3 – 0 – 0	03	V
216H05L501	Application Development Lab II - Android – Kotlin / Flutter	0 – 2 – 0	02	0 – 1 – 0	01	V
216H05C601	Application development using React Native	3 – 0 – 0	03	3 – 0 – 0	03	VI
216H05T601	Application Development Lab III - React Native	0 – 0 – 2	02	0 – 0 – 2	02	VI
216H05C701	Product Development material design (Swift)	3 – 0 – 0	03	3 – 0 – 0	03	VII
216H05T701	Application Development Lab IV - Product Development material design (Swift & iOS)	0 – 0 – 2	02	0 – 0 – 2	02	VII
	Total	12 – 4 – 4	20	12 – 2 – 4	18	

****All courses are project/mini-project based no separate project needed**

Proposed Examination Scheme

Course Code	Course Name	Examination Scheme				
		Marks				
		CA		ESE ^s	Lab/ Tut CA	Total
		ISE ^s	IA			
216H05C401	Fundamentals of Swift	30	20	50	--	100
216H05L401	Application Development Lab I - Swift	-	-	-	50	50
216H05C501	Mobile application development using Android – Kotlin / Flutter	30	20	50	--	100
216H05L501	Application Development Lab II - Android – Kotlin / Flutter	-	-	-	50	50
216H05C601	Application development using React Native	30	20	50	--	100
216H05T601	Application Development Lab III - React Native	-	-	-	75	75
216H05C701	Product Development material design (Swift)	30	20	50	--	100
216H05T701	Application Development Lab IV - Product Development material design (Swift & iOS)	-	-	-	75	75
Total		120	80	200	250	650

\$Onscreen Examination

Course Code	Name of the Course				
216H05C401	Fundamentals of Swift				
Teaching Scheme (Hrs./Week)	TH	P	TUT	Total	
	03	-	-	03	
Credits Assigned	03	-	-	03	
Evaluation Scheme	Marks				
	LAB/TUT CA	CA (TH)		ESE	Total
		IA	ISE		
	-	20	30	50	100

Course pre-requisites: Fundamentals of Programming language
Course Objectives:

1. Master Swift language syntax and advanced features.
2. Apply object-oriented programming and advanced Swift concepts.
3. Develop efficient applications with RxSwift, caching, and encryption.

Course Outcomes (CO):

1. Demonstrate a clear understanding of the Swift programming language and its syntax.
2. Implement Swift protocols, functions, and extensions for advanced app development.
3. Implement Swift object-oriented programming, including classes, type casting, and generics.
4. Apply advanced Swift concepts, including JSON, operators, closures, and concurrency.
5. Develop efficient Swift applications with RxSwift, caching, encryption, and performance

Module No.	Unit No.	Contents	No of Hrs.	CO
1	Introduction to Swift and Playgrounds		08	CO1
	1.1	Variables & Properties, Numbers, Strings and Characters, Booleans		
	1.2	Arrays, Tuples, Enums, Structs, Sets, Dictionaries		
	1.3	Switch, Optional, Conditionals, Error Handling, Loops		
	1.4	Xcode: Building, Running, Debugging, Interface Builder Basics Introduction to UIKit ,Display Data, Controls – Action		
2	Swift Protocols, functions and Extension		08	CO2
	2.1	Protocols :Protocol Basics, Delegate pattern, Associated type requirements, Class-Only Protocols, Protocol extension for a specific conforming class, Using the RawRepresentable protocol (Extensible Enum), Implementing Hashable protocol		
	2.2	Functions: Functions with Parameters, Subscripts, Methods , Variadic Parameters, Operators are Functions , Passing and returning functions Function types, Inout Parameters ,Throwing Error, Returning Values ,Trailing Closure Syntax,Functions With Closures		
	2.3	Extensions, Variables and functions, Initializers in Extensions, Subscripts, Protocol extensions, Restrictions		
3	Object Oriented Swift		08	CO3
	3.1	Classes: Defining a Class, Properties and Methods ,Reference ,Semantics, Classes and Multiple Inheritance deinit		
	3.2	Type Casting: Downcasting, Type casting in Swift Language ,Upcasting ,Example of using a downcast on a function parameter involving subclassing ,Casting with switch		
	3.3	Generics: The Basics of Generics, Constraining Generic Placeholder Types , Generic Class Examples , Using Generics to Simplify Array Functions, Advanced Type Constraints ,Generic Class Inheritance , Use generics to enhance type-safety OptionSet: OptionSet Protocol		
4	Advanced Swift		10	CO4
	4.1	Reading & Writing JSON, Advanced Operators (Custom Operators, Overflow, Commutative, Overloading + for Dictionaries)		
	4.2	Method Swizzling, Reflection, Access Control		
	4.3	Closures, Initializers, Concurrency, Functional Programming in Swift, Blocks(Escaping and non-Escaping), The Defer Statement, NSRegularExpression,		
5	Swift Application Management		11	CO5
	5.1	RxSwift, Swift Package Manager, Caching on disk space		
	5.2	Completion Handler, Buffer Pointers		
	5.3	Cryptographic Hashing, AES encryption, PBKDF2 Key Derivation, Logging in Swift, Memory Management, Performance		
Total			45	--

Reference Books*

Sr. No	Name/s of Author/s	Title of Book	Publisher	Edition/Year
1	Ahmad Sahar	iOS 14 Programming for Beginners	Packt Publisher	Fifth edition 2020
2	Jesse Feiler	Learn Computer Science with Swift	Apress	First 2018
3	Arpit Kulsreshtha	iOS 15 Application Development for Beginners	BPB Publications	First 2021
4*	https://books.goalkicker.com/SwiftBook/			
5*	https://developer.apple.com/swift/resources/			

*In addition to printed books, faculty can suggest (authentic) URLs or e-books, e-contents etc.

Course Code	Name of the Course			
216H05L401	Application Development Lab I - Swift			
Teaching Scheme (Hrs./Week)	TH	P	TUT	Total
	--	02	--	02
Credits Assigned	--	01	--	01
Evaluation Scheme	Marks			
	LAB/TUT CA*	CA (TH)		Total
		IA	ISE	
	50	--	--	50

LAB/TUT CA:

*LAB/TUT CA is an evaluation carried out during the said laboratory/tutorial throughout the semester on a continuous basis. In case of Laboratory, it can be a combination of laboratory experiments performed (at least 8-10), written record of experiments (Journal), Viva/On-screen test and/or Quiz, programming assignments (wherever applicable) and practical examination (if any) conducted during the semester. In case of tutorial, it can be a combination of graded assignments, group/individual activities such as presentations, group discussion, report writing etc. (as applicable).

Please note:

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2. Course coordinator should decide the rubrics/distribution of marks for different components in consultation with all other faculty teaching the same course.
3. The rubrics/distribution should be uniform for all batches.
4. The rubrics/distribution should be communicated to all students at the beginning of the semester.

Course Code	Name of the Course				
216H05C501	Mobile application development using Android – Kotlin				
Teaching Scheme (Hrs./Week)	TH	P	TUT	Total	
	03	--	--	03	
Credits Assigned	03			03	
Evaluation Scheme	Marks				
	LAB/TUT CA	CA (TH)		ESE	Total
		IA	ISE		
	--	20	30	50	100

Course pre-requisites: Basic of Java programming
Course Objectives:

To build a native mobile Application for Android platform using Kotlin and deploy it on Google Playstore.

Course Outcomes (CO):

1. Understand basics, object-oriented programming concepts and Null safety in Kotlin.
2. Understand Android studio, its project structure and Android UI design.
3. Implement UI components, Widgets and Navigation using Kotlin.
4. Connect Mobile Application with external API and Databases.
5. Incorporate App monetisation, build and deploy Mobile App on Google Playstore.

Module No.	Unit No.	Contents	No of Hrs.	CO
1	Introduction to Kotlin Programming		8	CO1
	1.1	Kotlin Variables, Data Types, Operators and Expressions		
	1.2	Kotlin Decisions and Loops		
	1.3	Kotlin Functions, Kotlin Lambdas, Higher-order functions		
	1.4	Object Oriented Programming: Classes, Objects, Constructors, Inheritance Advanced OOP Concepts: Interface, Abstract Class, Generic Classes Null Safety in Kotlin		
	1.5	Collections, Asynchronous Programming and Coroutines		
2	Introduction to Android Studio and Project Structure		7	CO2
	2.1	Installing the Android Studio, Exploring Android emulator and Setup Kotlin and Run First App		
	2.2	Android Studio Tour – Project Structure, File Structure, Manifest, Strings		
	2.3	Introduction to Android UI design and Material Design		
	2.4	Android Lifecycle		
3	User Interface and Navigation in Kotlin		12	CO3
	3.1	Activities Views, layouts and Common UI components, Creating UI through code and XML		
	3.2	Introduction to Widgets: TextViews, EditText and Buttons, ImageView, Seekbar, ListView, RecyclerView, ActionBar, SanckBar, WebView, Canvas, GridView, ListView, Spinner		
	3.3	Introduction to Jetpack Compose		
	3.4	Display lists and use Material Design		
	3.5	Interacting with UI and state		
	3.6	Activities, Navigation and app Architecture		
4	Connecting to the API and Backend		12	CO4
	4.1	Introduction to networking connection		
	4.2	Using GPS and Google Maps		
	4.3	Multimedia, Sensor and Services		
	4.4	Connecting and Consuming JSON data and API from the internet		
	4.5	Connecting to the Firebase database		
5	App Monetization & Build and Deploy App to Playstore		6	CO5
	5.1	Including Ads, Implementing Banner and Interstitial Ads		
	5.2	App Monetization Methods		
	5.3	Creating signed .apk, App Bundles		
	5.4	Going Live on Playstore		
Total			45	

Reference Books*

Sr. No	Name/s of Author/s	Title of Book	Publisher	Edition/Year
1	John Horton	Android Programming with Kotlin for Beginners	Packt Publication	2019
2	Cataline Ghita	Kickstart Modern Android Development with Jetpack and Kotlin	Packt Publication	2022
3	Dawn Griffiths, David Griffiths	Head First Android Development: A Learner's Guide to Building Android Apps with Kotlin	O'Reilly Media Inc.	3 rd Edition, 2022
4	Alex Forrester, Eran Boudjnah, Alexandru Dumbravan, Jomar Tigcal	How To Build Android Apps With Kotlin	Packt Publication	2 nd Edition, 2023
5	Aanand Shekhar Roy, Rashi Karanpuria	Kotlin Programming Cookbook	Packt Publication	2018

*In addition to printed books, faculty can suggest (authentic) urls or e-books, e-contents etc.

Course Code	Name of the Course				
216H05C501	Mobile application development using Android –Flutter				
Teaching Scheme (Hrs./Week)	TH	P	TUT	Total	
	03	--	--	03	
Credits Assigned	03	--	--	03	
Evaluation Scheme	Marks				
	LAB/TUT CA	CA (TH)		ESE	Total
		IA	ISE		
		--	20	30	50

Course pre-requisites: Basic understanding of programming

Course Objectives:

To Build real native mobile apps for the two most popular mobile platforms - iOS and Android deploy it on Google Playstore/iOS App Store.

Course Outcomes (CO):

1. Understand basics, object-oriented programming concepts and Null safety in Dart & Flutter.
2. Configure system for Flutter Development and Understand Android studio, Android UI design.
3. Implement UI components, Widgets and Navigation using Flutter.
4. Connect Mobile Application with external API and Firebase Database.
5. Understand state Management and Bloc Pattern.
6. Incorporate App monetisation, build and deploy Mobile App on Google Playstore/ iOS App Store.

Module No.	Unit No.	Contents	No of Hrs.	CO
1	Introduction to development with flutter		8	CO1
	1.1	Introduction to Dart Programming Language: Coding style and naming convention, Declaring variables, Numbers, Booleans, const and final keywords Dart Types, Dart operators, Control flow		
	1.2	Dart Functions, Lambdas, Higher-order functions		
	1.3	Object Oriented Programming: Classes, Objects, Constructors, Inheritance Advanced OOP Concepts: Interface, Abstract Class, Generic Classes Null Safety in Flutter		
	1.4	Collections & Asynchronous Programming		
2	Introduction to Flutter Framework, Android Studio and the Project Structure		5	CO2
	2.1	Setting up Development Environment: Understanding the Flutter Architecture, Flutter macOS Setup, Flutter Windows setup.		
	2.2	Installing the Android Studio, Exploring Android emulator and Setup Flutter and Run First App		
	2.3	Introduction to Android UI design and Material Design		
3	User Interface and Navigation in Flutter		12	CO3
	3.1	Basics and Understanding Widgets: Exploring default project structure, Flutter Basic Layouts, Widgets in flutter, Adding custom images, fonts.		
	3.2	Working with TextFields, Buttons, Scaffold, Material App, Drawer, Navigation Bar, Floating Action button, Snack bar, Lists and Conditionals, Handling user input.		
	3.3	Passing Data via the Constructor, Using Named Routes & Passing Data with Named Routes, Screens & Navigation, onGenerateRoute & onUnknownRoute		
	3.4	Forms and validation with Flutter: Labels and Hint Text, Customizing Keyboard Type, Handling Password Inputs, Displaying Buttons with ElevatedButton, Changing Widget Colors, Layout Control, Form Validation, Referencing Widgets with Global Keys, The Form Widget and FormState, Creating a Global Key, Referencing FormState with Global Keys, Validating via FormState, Triggering Validation, Retrieving Form Values.		
4	Connecting to the API and Backend		8	CO4
	4.1	HTTP Requests with Flutter Photos API, Working with JSON, The HTTP Package.		
	4.2	Firebase console, Firebase services - Authentication, Realtime Database.		
5	State Management		7	CO5

	5.1	What is State and How do we Manage it, How do Callbacks work in Dart? The ListView Builder, Lifting State Up Challenge, Flutter App Architecture Patterns. Introducing the Provider Package, Using a Provider and a Change Notifier to Manage State.		
	5.2	Advanced Flutter Architecture – The Bloc Pattern BLOC's vs Stateful Widgets, The Purpose of Streams with Blocs, Annotating Stream Types, and Issues with Bloc Access.		
6	App Monetization & Build and Deploy App to Playstore/ App Store		5	CO6
	6.1	App Monetization Methods		
	6.2	Deploying flutter App on Playstore		
	6.3	Deploying flutter App on App Store		
Total			45	--

Reference Books*

Sr. No	Name/s of Author/s	Title of Book	Publisher	Edition/Year
1	Eric Windmill	Flutter in Action	Manning Publications	1 st Edition, 2019
2	Marco L. Napoli	Beginning Flutter: A Hands On Guide to App Development	Wrox	1 st Edition
3	Prajyot Mainkar , Salvatore Giordano	Google Flutter Mobile Development Quick Start Guide	Packt Publishing	1 st Edition, 2019
4	Ivo Balbaert , Dzenan Ridjanovic	Learning Dart	Packt Publishing	2 nd Edition, 2015
5	Mark Clow	Learn Google Flutter Fast: 65 Example Apps	Independently published	1 st Edition, 2019

*In addition to printed books, faculty can suggest (authentic) URL's or e-books, e-contents etc.

Course Code	Name of the Course			
216H05L501	Application Development Lab II - Android – Kotlin / Flutter			
Teaching Scheme (Hrs./Week)	TH	P	TUT	Total
	--	02	--	02
Credits Assigned	--	01	--	01
Evaluation Scheme	Marks			
	LAB/TUT CA*	CA (TH)		Total
		IA	ISE	
	50	--	--	50

LAB/TUT CA:

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Course Code	Name of the Course				
216H05C601	Application development using React Native				
Teaching Scheme (Hrs./Week)	TH	P	TUT	Total	
	03	-	-	03	
Credits Assigned	03	-	-	03	
Evaluation Scheme	Marks				
	LAB/TUT CA	CA (TH)		ESE	Total
		IA	ISE		
	-	20	30	50	100

Course pre-requisites: Fundamentals of JavaScript.

Course Objectives:

The objective of the proposed React Native course is to equip learners with the knowledge and skills needed to develop mobile applications using the React Native framework.

1. Understanding React Native Basics.
2. React and React Native Concepts.
3. React Native Components and Styling.
4. Networking and Data Management.

Course Outcomes (CO):

1. Demonstrate Proficiency in React Native Development and Implement Navigation and State Management.
2. Apply Data Management and Networking Techniques.
3. Apply concepts to enhance the functionality and user experience of their React Native applications.
4. Develop and Deploy Real-world React Native Applications.

Module No.	Unit No.	Contents	No of Hrs.	CO
1	Introduction to React Basics		12	CO1
	1.1	Introduction to React: • Components and Props • State and Lifecycle • Handling Events		
	1.2	JSX and Component Composition: • Understanding JSX syntax • Component composition and reusability		
	1.3	React Native Fundamentals: React Native Components: • Core components (View, Text, Image, etc.) • Styling in React Native • Flexbox layout in React Native		
	1.4	Navigation in React Native: • React Navigation library • Stack Navigator, Tab Navigator, Drawer Navigator		
2	State Management, Networking and Data Management		10	CO2
	2.1	State Management in React Native • Using local state (useState hook). • Implementing useReducer for state management. • Introduction to the Context API.		
	2.2	Redux in React Native • Understanding the principles of Redux. • Integrating Redux for global state management.		
	2.3	Fetching Data: • Making API requests using Fetch or Axios. • Handling asynchronous operations with Promises or async/await. Data Storage: • Utilizing AsyncStorage for local data storage. • Exploring other local storage options (SQLite, etc.).		
3	Advanced Topics: Animations in React Native		8	CO3
	3.1	Implementing animations using the React Native Animated API. Applying animations to enhance user interfaces.		
	3.2	Native Modules and Device Features: • Integrating native modules for advanced functionality. • Accessing device features such as the camera, GPS, etc		
4	Testing and Debugging and Deployment		15	CO4

	4.1	Testing React Native Applications: • Writing unit tests with Jest. • Conducting UI tests with Detox.		
	4.2	Debugging Techniques: • Using React DevTools for debugging. • Debugging on emulators/simulators and real devices.		
	4.3	Deployment: • Building applications for iOS and Android. • Guidelines for app store deployment.		
Total			45	--

Reference Books*

Sr. No	Name/s of Author/s	Title of Book	Publisher	Edition/Year
1	Bonnie Eisenman	Learning React Native Building Native Mobile Apps with JavaScript	O'Reilly Media	2 nd / 2017
2	Eric Masiello, Jacob Friedmann	Mastering React Native	Packt Publishing	2017
3	Devin Abbott, Houssein Djirdeh, Anthony Accomazzo, Sophia Shoemaker	Fullstack React Native Create beautiful mobile apps with JavaScript and React Native	Independently published	2019
4	https://reactnative.dev/docs/getting-started			
5	https://reactnavigation.org/docs/getting-started/			
6	https://jestjs.io/docs/getting-started			

*In addition to printed books, faculty can suggest (authentic) URLs or e-books, e-contents etc.

Course Code	Name of the Course			
216H05T601	Application Development Lab III - React Native			
Teaching Scheme (Hrs./Week)	TH	P	TUT	Total
	--	--	02	02
Credits Assigned	--	--	02	02
Evaluation Scheme	Marks			
	LAB/TUT CA*	CA (TH)		Total
		IA	ISE	
	75	--	--	75

LAB/TUT CA:

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Course Code	Name of the Course				
216H05C701	Product Development material design (Swift)				
Teaching Scheme (Hrs./Week)	TH	P	TUT	Total	
	03	--	-	03	
Credits Assigned	03	--	-	03	
Evaluation Scheme	Marks				
	LAB/TUT CA	CA (TH)		ESE	Total
		IA	ISE		
	--	20	30	50	100

Course pre-requisites: Fundamentals of Programming language, Swift Programming
Course Objectives:

1. Master the principles of Material Design, focusing on responsive interaction, hierarchy, color schemes, and animations in Swift.
2. Implement the Model-View-Controller (MVC) design pattern in iOS applications.
3. Utilize controllers, classes, and model objects to structure and manage data.
4. Design and organize code effectively to create modular and maintainable iOS applications.
5. Configure continuous integration and delivery workflows using Xcode Cloud.
6. Implement secure REST API calls in Swift for data retrieval and manipulation.
7. Troubleshoot and optimize Xcode Cloud builds, ensuring a smooth development and deployment process.

Course Outcomes (CO):

1. Develop Proficiency in Swift and Material Design Principles
2. Apply Architectural and Design Patterns in iOS Development
3. Design and Implement, Integrate Xcode Cloud and REST API

Module No.	Unit No.	Contents	No of Hrs.	CO
1	Material Design Principles 1		08	CO1 CO2
	1.1	Responsive Interaction, Hierarchy And Scale, Consistency Across Devices, Color And Theming, Depth And Shadows, Animations And Transitions		
	1.2	Swift UI Components And Material Design: Buttons And Action Items, Text Fields, Cards And Containers, Lists And Navigation, Customizing Appearance		
	1.3	Implementing Material Design Color Schemes In Swift: Primary And Secondary Colors, Color Variations, Error And Warning Colors Using Color Assets, Adapting To Dark Mode		
	1.4	Adapting Swift Animations With Material Motion: Understanding Material Motion, Shared Element Transitions, Responsiveness And Feedback, Natural Motion Paths, Interruptible Animations, Sequencing And Choreography		
2	Material Design Principles 2		08	CO1 CO2
	2.1	Best Practices For Swift Material Design: Consistency Is Key Opt For Modular Components, Prioritize Responsiveness, Stay Updated With Design Guidelines, Test Across Devices, Accessibility Matters		
	2.2	Common Pitfalls And Solutions: Overloading UI With Animations Inconsistent Implementation Across iOS Devices, Ignoring Accessibility, Overcomplicating UI Components, Not Keeping Up With Updates, Not Handling Failures Gracefully		
3	Architecture and design principle in action		08	CO1 CO2
	3.1	Understanding the Model-View-Controller design pattern		
	3.2	Exploring controllers and classes		
	3.3	Understanding model objects Understanding .plist files Creating a structure to represent a cuisine Implementing a data manager class to read data from a .plist file Displaying data in a collection view		
	3.4	Understanding and creating annotations Adding annotations to a map view Organizing your code		
4	Working with Image and widgets		10	CO1 CO2
	4.1	Creating model objects for the Photo Filter screen Creating the ImageFiltering protocol Creating view controllers for the Photo Filter screen		
	4.2	Introducing widgets Adding a widget target to your app Providing timeline entries to your widget		

		Customizing your widget's view Adding a widget to your Home Screen		
	4.3	Introducing app clips Adding an app clip to your project Configuring your app clip		
		Xcode Cloud and Rest API		
	5.1	About continuous integration and delivery with Xcode Cloud Requirements for using Xcode Cloud Setup and maintenance Configuring your first Xcode Cloud workflow Making dependencies available to Xcode Cloud Configuring Xcode Cloud for your team Building Swift packages and Swift Playgrounds app projects with Xcode Cloud Setting the next build number for Xcode Cloud builds Including notes for testers with a beta release of your app Removing your project from Xcode Cloud		
5	5.2	Usage data Reviewing Xcode Cloud usage data Workflows Developing a workflow strategy for Xcode Cloud Xcode Cloud workflow reference Creating a workflow that builds your app for distribution Source code management Source code management setup Configuring requirements for merging a pull request Custom build scripts Writing custom build scripts Environment variable reference Troubleshooting Resolving common configuration and build issues Resolve GitHub Enterprise connection issues Reporting feedback for Xcode Cloud Notifications Configuring webhooks in Xcode Cloud Connecting Xcode Cloud to Slack	11	CO3
	5.3	REST API REST API Call in Swift Integrating authentication and response handling Calling REST APIs from within an iOS app in SwiftUI Handling HTTP Methods And Responses Optimizing API Requests Protocol-oriented networking layer architecture in Swift Securing Your Swift REST API		
Total			45	--

Reference books:

Sr. No	Name/s of Author/s	Title of Book	Publisher	Edition/Year
1	Ahmad Sahar	iOS 14 Programming for Beginners	Packt Publisher	Fifth edition 2020
2	Jesse Feiler	Learn Computer Science with Swift	Apress	First 2018
3	Arpit Kulsreshtha	iOS 15 Application Development for Beginners	BPB Publications	First 2021
4*	https://books.goalkicker.com/SwiftBook/			
5*	https://developer.apple.com/swift/resources/			
6*	https://developer.apple.com/documentation/appstoreconnectapi/xcode_cloud_workflows_and_builds/			
7*	https://marketsplash.com/tutorials/swift/swift-rest-api/			

*In addition to printed books, faculty can suggest (authentic) URLs or e-books, e-contents etc.

Course Code	Name of the Course				
216H05T701	Application Development Lab IV - Product Development material design (Swift & iOS)				
Teaching Scheme (Hrs./Week)	TH	P	TUT	Total	
	--	--	02	02	
Credits Assigned	--	--	02	02	
Evaluation Scheme	Marks				
	LAB/TUT CA*	CA (TH)		ESE	Total
		IA	ISE		
	75	--	--	--	75

LAB/TUT CA:

***LAB/TUT CA is an evaluation carried out during the said laboratory/tutorial throughout the semester on a continuous basis. In case of Laboratory, it can be a combination of laboratory experiments performed (at least 8-10), written record of experiments (Journal), Viva/On-screen test and/or Quiz, programming assignments (wherever applicable) and practical examination (if any) conducted during the semester. In case of tutorial, it can be a combination of graded assignments, group/individual activities such as presentations, group discussion, report writing etc. (as applicable).**

Please note:

13. The total marks assigned for Continuous Assessment (LAB/TUT) as per scheme can be distributed in a number of components as given above.
14. Course coordinator should decide the rubrics/distribution of marks for different components in consultation with all other faculty teaching the same course.
15. The rubrics/distribution should be uniform for all batches.
16. The rubrics/distribution should be communicated to all students at the beginning of the semester.